Black Holes in Darke Theory

Confinement, Structural Admissibility, and the Coherence Boundary

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Abstract

We develop a standalone Darke-theoretic framework for black holes by unifying three layers of structure: confinement, coherence burden, and the geometric coherence boundary. In this formulation, a black hole is not treated primarily as a geometric singularity object, but as a terminal confinement regime whose persistence and external recoverability are governed by structural admissibility. The event horizon is reinterpreted as the locus at which the coherence burden required for ordinary exterior geometric recoverability exceeds the admissible support budget of the governing system. The interior is correspondingly reinterpreted not as a literal singularity, but as a maximal closure-dominant confinement phase in which classical descriptive variables cease to remain structurally complete.

We then incorporate a strict coherence-boundary evaporation sector. Geometry is taken to remain valid only when a coherence functional satisfies $\Phi \geq 1$, with $\Phi(M) = M/M_P$. Entropy is interpreted as the logarithm of accessible geometric phase space, and smooth first-order collapse of this phase space at the Planck threshold yields a modified entropy law, a finite threshold temperature, and a non-divergent evaporation equation. Radiation terminates not because temperature vanishes, but because no admissible lower-mass geometric microstates exist below the boundary. The resulting endpoint is a terminal Planck-scale boundary state reached in finite time. The overall picture therefore replaces the orthodox singular and divergent endpoint with a structurally admissible confinement-boundary termination.

1 Introduction

Black holes remain among the most successful and most problematic objects in modern theoretical physics. Their external phenomenology is strongly supported, yet the standard account inherits three persistent conceptual tensions. First, continued classical extrapolation into the interior yields singularity. Secondly, semiclassical evaporation leads to the information problem. Thirdly, the standard Hawking temperature diverges as $M \to 0$, producing an endpoint pathology unless additional physics is imposed.

Darke Theory suggests a different route into these questions. In the broader Darke lineage, persistent structure is not primitive. It emerges downstream of recursive differentiation, closure, and admissibility. Structure exists not merely because it can be formally described, nor merely because it can appear transiently, but because it can be carried stably within the support conditions of the governing system. The Coherence Burden Principle makes this regime-level logic explicit: coherence is the currency of persistence, support is finite or bounded, lifetime matters as much as local burden, and admissibility functions as a regime filter rather than a local yes-or-no label. A candidate regime is structurally admissible only if the coherence required for its persistence can be supported for the necessary lifetime within the closure, refinement, and constraint structure of the governing system.

This regime logic is directly relevant to black holes. Collapse may be reinterpreted as the progressive growth of confinement burden. The key question is no longer merely whether geodesics escape, but whether the system can still support the coherence burden required for recoverable exterior geometric structure. Once that burden exceeds admissible support, horizon

formation follows as a structural transition. The singularity then ceases to be the primary ontology and instead becomes a marker that classical description has been pushed beyond its admissible regime.

A second Darke-theoretic ingredient sharpens the endpoint further. In the coherence-boundary evaporation construction, geometry is treated as valid only when a coherence functional Φ satisfies $\Phi \geq 1$, with $\Phi(M) = M/M_P$. Entropy is interpreted statistically as the logarithm of accessible geometric phase-space volume. Enforcing smooth first-order collapse of that accessible phase space at the Planck threshold yields a modified entropy law, a finite threshold temperature, and an evaporation equation that terminates at $M = M_P$ in finite time without divergence.

The purpose of the present paper is to unify these strands into one standalone black-hole framework. The paper argues that:

1. a black hole is a terminal confinement regime rather than primarily a singularity object,
2. the horizon is the boundary at which the coherence burden of exterior geometric recoverability exceeds admissible support,
3. the interior is a closure-dominant maximal confinement phase rather than a literal physical infinity,
4. evaporation remains meaningful only within the geometric coherence regime,
5. the endpoint is a finite coherence-boundary termination rather than a divergent disappearance.

The result is a Darke-theoretic black-hole picture in which confinement, burden, closure, and geometric validity all sit inside one structurally continuous account.

2 Foundational Setting

The present construction stands downstream of the core Darke-theoretic sequence

$$\text{recursive differentiation} \implies \text{closure} \implies \text{admissibility} \implies \text{persistent structure.}$$

This compressed sequence is not offered as a substitute for the foundational papers, but as the minimum load-bearing lineage needed for the present argument. Recursive differentiation generates structural possibility but not yet stable structure. Closure stabilises selected patterns against dispersion, but closure alone does not guarantee indefinite persistence. Admissibility then enters as the decisive filter: can the branch be borne without contradiction, fragmentation, unsupported renewal cost, or regime-level structural failure?

The Coherence Burden Principle sharpens this sequence at the regime level. It explains that support and lifetime are the operational variables through which closure and admissibility become visible. If closure is what allows a structure to stabilise, and admissibility is what determines whether such stabilisation is genuinely supportable, then coherence burden is the practical currency in which that question is paid.

This immediately suggests a new reading of gravitational collapse. A black hole is not just the place where a metric coefficient changes sign or where null escape fails. It is the regime reached when the coherence required for the persistence of ordinary exterior geometric recoverability can no longer be carried by the system. Collapse therefore becomes a burden problem as much as a geometry problem.

3 Operational Variables

We now define the burden and support fields explicitly at the regime level.

Let $\rho(x,t)$ denote the local mass-energy density. Let $K(x,t) \geq 0$ denote a positive scalar curvature burden, taken for example to be a normalised Kretschmann-type invariant or any equivalent non-negative curvature scalar appropriate to the effective description. Let $\theta(x,t)$ denote the local expansion scalar of the collapsing flow, and define the compressive component

$$\Theta_-(x,t) := \max\{0, -\theta(x,t)\},$$

so that $\Theta_- > 0$ only in genuinely convergent regions.

To keep the construction dimensionless and structurally transparent, introduce reference scales ρ_*, K_*, and Θ_*, and define the normalised fields

$$\hat{\rho}(x,t) := \frac{\rho(x,t)}{\rho_*}, \qquad \hat{K}(x,t) := \frac{K(x,t)}{K_*}, \qquad \hat{\Theta}(x,t) := \frac{\Theta_-(x,t)}{\Theta_*}.$$

Definition 1 (Burden density)**.** *The local coherence-burden density is defined by*

$$b(x,t) := \alpha\,\hat{\rho}(x,t) + \beta\,\hat{K}(x,t) + \gamma\,\hat{\Theta}(x,t),$$

where $\alpha, \beta, \gamma > 0$ are weighting constants.

This definition expresses the idea that burden rises with increasing density, increasing curvature load, and increasing inward renewal/compression demand. These are exactly the local contributions expected to raise the cost of sustaining recoverable exterior geometry during collapse.

We now define the support side. Let

$$\Phi_{\mathrm{loc}}(x,t)$$

denote a local geometric coherence factor, interpreted as the residual capacity of the geometry at (x,t) to sustain admissible exterior recoverability. The minimal requirement for the present framework is that Φ_{loc} be non-negative and decrease as confinement, curvature burden, and compressive collapse intensify. We take the simplest compatible form

$$\Phi_{\mathrm{loc}}(x,t) := \frac{1}{1 + b(x,t)}.$$

Definition 2 (Support density)**.** *The local support density is defined by*

$$s(x,t) := \Phi_{\mathrm{loc}}(x,t) = \frac{1}{1 + b(x,t)}.$$

This gives a direct and closed burden-support pair. As collapse intensifies, $b(x,t)$ increases and $s(x,t)$ correspondingly decreases. The support field therefore encodes the remaining local carrying capacity of the geometric regime.

Definition 3 (Persistence horizon)**.** *Let $\tau > 0$ denote the required persistence horizon of the regime. In different settings τ may be temporal, scale-depth, cascade length, or another domain-specific lifetime variable. What matters is that admissibility is judged across a horizon rather than at a single local instant.*

Definition 4 (Integrated support and burden)**.** *For a spacetime region $\mathcal{R} \subset \Sigma \times [0,\tau]$, define*

$$\mathcal{S}(\mathcal{R};\tau) := \int_0^\tau \int_{\mathcal{R}(t)} s(x,t)\, dV_t\, dt,$$

and

$$\mathcal{B}(\mathcal{R};\tau) := \int_0^\tau \int_{\mathcal{R}(t)} b(x,t)\, dV_t\, dt.$$

Definition 5 (Confinement functional)**.** *The confinement functional of the regime over* $(\mathcal{R}, \tau)$ *is*

$$\Lambda(\mathcal{R};\tau) := \frac{\mathcal{B}(\mathcal{R};\tau)}{\mathcal{S}(\mathcal{R};\tau)}, \qquad \mathcal{S}(\mathcal{R};\tau) > 0.$$

Substituting the explicit support definition yields

$$\Lambda(\mathcal{R};\tau) = \frac{\displaystyle\int_0^\tau \int_{\mathcal{R}(t)} b(x,t)\, dV_t\, dt}{\displaystyle\int_0^\tau \int_{\mathcal{R}(t)} \frac{1}{1+b(x,t)}\, dV_t\, dt}.$$

This form makes the formation criterion fully closed at the regime-functional level. The same local structural loads that raise confinement burden simultaneously reduce local support, so collapse drives Λ upward in a mathematically explicit way.

Remark 1. *The choice*

$$s = \frac{1}{1+b}$$

is the minimal monotone closure compatible with the burden principle. More elaborate support laws could be introduced in later work, but the present form already captures the required structure: increasing density, curvature burden, and compressive demand raise the local cost of recoverable geometry while reducing the remaining admissible support budget.

4 Black Holes as Terminal Confinement Regimes

We now state the central ontological claim.

Definition 6 (Terminal confinement regime)**.** *A black hole is a terminal confinement regime in which closure becomes locally dominant, outward structural recoverability fails, and the system transitions into a self-trapped phase whose persistence is governed by admissibility rather than by unrestricted geometric extrapolation.*

This definition deliberately changes the explanatory order. In the orthodox reading, the black hole is first a strong-gravity object, then a horizon object, then potentially a singular one. In the present reading, the black hole is first a confinement object. Strong-gravity phenomenology is the external manifestation of that confinement, not its deepest description.

Let B_r denote the closed ball of areal radius r centred on the collapse region at a fixed time slice, and let τ_r denote the persistence horizon required for the regime inside B_r to count as a recoverable exterior geometric branch. Then the confinement profile is

$$\Lambda(r) := \Lambda(B_r;\tau_r) = \frac{\int_0^{\tau_r} \int_{B_r(t)} b(x,t)\, dV_t\, dt}{\int_0^{\tau_r} \int_{B_r(t)} s(x,t)\, dV_t\, dt}.$$

Definition 7 (Critical confinement threshold)**.** *A critical confinement threshold is a value* Λ_c*, taken here to be normalised to*

$$\Lambda_c = 1,$$

such that recoverable exterior geometry remains structurally admissible for

$$\Lambda(r) < 1$$

and fails for

$$\Lambda(r) \geq 1.$$

This normalisation simply fixes units so that the admissibility boundary occurs when required burden equals available support. Any alternative threshold value can be absorbed by rescaling b or s.

Principle 1 (Black-hole formation criterion)**.** *A bounded collapse region forms a black hole if there exists a radius r_h such that*

$$\Lambda(r_h) = 1$$

and

$$\Lambda(r) > 1 \quad \textit{for all} \quad 0 \leq r < r_h.$$

The smallest such r_h is the structural horizon radius.

This criterion states that the horizon is the minimal radius at which the burden-support ratio saturates admissibility. Interior to that radius, the system can still exist, but no longer as an ordinarily recoverable geometric regime.

5 The Horizon as a Structural Admissibility Boundary

We can now define the horizon more sharply.

Definition 8 (Structural horizon)**.** *The structural horizon is the minimal radius*

$$r_h := \inf\{r > 0 : \Lambda(r) \leq 1\}$$

such that

$$\Lambda(r) > 1 \quad \textit{for } r < r_h, \qquad \Lambda(r_h) = 1.$$

Equivalently, if one defines the exterior recoverability margin

$$\mathcal{R}(r) := 1 - \Lambda(r),$$

then

$$\mathcal{R}(r) > 0$$

corresponds to supportable exterior recoverability,

$$\mathcal{R}(r) = 0$$

defines the horizon, and

$$\mathcal{R}(r) < 0$$

marks the closure-dominant interior.

This is more informative than saying only that signals cannot escape. The point is not merely that trajectories fail to re-emerge. The point is that the system can no longer support the persistence conditions required for its internal organisation to remain recoverable in the ordinary exterior geometric basis.

The collapse sequence may now be written as:

1. Compression intensifies and internal refinement density rises.
2. The integrated burden $\mathcal{B}(B_r; \tau_r)$ grows faster than the integrated support $\mathcal{S}(B_r; \tau_r)$.
3. The margin $\mathcal{R}(r) = 1 - \Lambda(r)$ decreases toward zero.
4. At $r = r_h$, one has $\mathcal{R}(r_h) = 0$.
5. For $r < r_h$, one has $\mathcal{R}(r) < 0$, and the regime becomes closure-dominant.

Remark 2. *This makes the horizon a regime boundary rather than merely a local kinematic feature. Admissibility is not a pointwise label attached to an isolated event, but a higher-level criterion evaluated over sustained realisability within closure-compatible support.*

6 The Interior: Maximal Closure Rather than Singularity

The singularity problem is now ready to be reframed. In the standard reading, continued extrapolation into the black-hole interior drives curvature scalars and densities toward divergence. In the present framework, this is interpreted not as evidence for a completed physical infinity, but as evidence that the classical descriptive basis has been driven beyond structural admissibility.

Definition 9 (Maximal closure state)**.** *A maximal closure state is a confinement phase in which*

$$\Lambda(r) > 1 \quad \textit{for all admissible interior radii } r < r_h,$$

and no refinement of the ordinary exterior geometric description restores

$$\Lambda(r) \leq 1$$

throughout the interior.

The singularity is therefore replaced by maximal closure. This is not a claim that the interior becomes simple, empty, or benign. It is a claim that classical divergence should be read as a breakdown signal rather than as a final ontology.

The gain is substantial. Once singularity is replaced by a closure-dominant regime boundary, several conceptual tensions soften at once:

1. divergence is no longer treated as a literal physical completion,
2. the interior remains physically meaningful without being classically describable in full,
3. information need not be annihilated merely because classical recoverability fails.

A regime may be writable while not being supportable as a stable branch in the originally chosen variable basis. Formal descriptibility and genuine supportability must not be confused.

7 Closure-Bound Information and Geometric Accessibility

The information problem may now be rephrased in more structurally precise terms. Once matter and field structure enter the confinement regime, the ordinary exterior basis ceases to remain adequate for full recovery. However, it does not follow that structure is destroyed absolutely. What follows is that the structure becomes closure-bound.

Definition 10 (Closure-bound information)**.** *Closure-bound information is structural content retained within a closure-dominant confinement regime but not externally recoverable in the ordinary coarse-grained geometric basis.*

This gives three informational layers:

1. **Exteriorly recoverable information:** structure available to ordinary exterior interrogation.
2. **Boundary-encoded information:** structure represented at or near the horizon as the accessible interface of the confined object.
3. **Closure-bound information:** structure retained in the maximal closure phase but inaccessible to ordinary exterior recovery.

The key refinement needed here is the following.

Principle 2 (Accessibility split)**.** *The collapse of accessible geometric phase space concerns only the exteriorly admissible geometric microstate sector. It does not imply erasure of closure-bound structural content, which lies outside the ordinary geometric accessibility counted by* Ω*.*

This principle is essential. When the paper later writes

$$S = k_B \ln \Omega,$$

the Ω in question counts admissible *geometric* microstates of the black-hole branch. It does not claim to count the entirety of closure-bound structural content. Hence phase-space collapse at the coherence boundary concerns the geometric branch and its radiative channels, not total ontological annihilation.

8 Coherence Boundary and Geometric Validity

The confinement framework above explains why black holes form and why the interior should not be treated as a literal singularity. We now require a second layer to govern the late evaporation regime. This is provided by the strict coherence boundary construction.

Define the Planck mass

$$M_P = \sqrt{\frac{\hbar c}{G}},$$

and introduce the dimensionless coherence functional

$$\Phi(M) = \frac{M}{M_P}.$$

Definition 11 (Geometric validity condition)**.** *The geometric black-hole regime is structurally admissible only when*

$$\Phi(M) \geq 1.$$

Equivalently, the geometric branch terminates at

$$M = M_P.$$

Below this threshold, geometric encoding is not structurally admissible.

This condition must be read carefully. It is not a claim that physical reality ceases below M_P. It is a claim that the *geometric black-hole regime* ceases to remain valid below that threshold. In the language of the present paper, the ordinary geometric description has reached its final admissible boundary.

The relation between Λ and Φ may now be stated more precisely. The confinement functional Λ measures admissibility at the level of regime formation: it determines whether a collapsing structure can still sustain recoverable exterior geometry over its persistence horizon. The coherence functional Φ, by contrast, measures admissibility at the terminal geometric scale of that same regime. In this sense, Φ is not an unrelated second principle, but the low-scale geometric projection of the same burden-support logic that Λ governs at formation.

The two quantities therefore operate at different stages of the black-hole life cycle while belonging to one common admissibility structure. The quantity Λ determines when the confinement regime forms, whereas Φ determines when its geometric branch can no longer be borne. Put differently, Λ governs formation-side admissibility and Φ governs termination-side geometric admissibility.

This relation may be summarised schematically as

$$\Lambda(r) = \frac{\mathcal{B}(B_r; \tau_r)}{\mathcal{S}(B_r; \tau_r)} \quad \Longrightarrow \quad \text{formation of the confinement regime,}$$

while

$$\Phi(M) = \frac{M}{M_P} \quad \Longrightarrow \quad \text{termination of the geometric black-hole branch at } M = M_P.$$

Thus the black hole is not born under one structural principle and terminated under another. It is born and terminated under one admissibility logic expressed at two different scales: first as a burden-support threshold of confinement, and later as a coherence threshold of the residual geometric branch.

9 Entropy as Accessible Geometric Phase Space

To describe the evaporation sector, we adopt the standard statistical interpretation

$$S = k_B \ln \Omega,$$

where $\Omega(M)$ denotes the number of accessible geometric microstates at mass M.

The classical Bekenstein entropy is

$$S_H(M) = \frac{k_B A}{4\ell_P^2} = \frac{4\pi k_B G M^2}{\hbar c},$$

with horizon area

$$A = \frac{16\pi G^2 M^2}{c^4}.$$

The key move is to reinterpret the Planck-threshold modification as a collapse in accessible geometric phase space. Write

$$\Omega_{DT}(M) = \Omega_H(M)\, f(M),$$

so that the modified entropy is determined by the admissible geometric state count rather than by a direct ad hoc deformation of temperature.

We impose three conditions:

1. classical quadratic scaling for $M \gg M_P$,
2. vanishing accessible geometric phase space at $M = M_P$,
3. smooth first-order collapse near threshold.

Let

$$\epsilon = M - M_P.$$

Smooth first-order collapse near threshold requires

$$\Omega_{DT}(M) \propto \epsilon \qquad \text{as } \epsilon \to 0^+.$$

To preserve classical M^2 scaling at large mass while enforcing linear vanishing at threshold, the minimal smooth polynomial deformation is

$$\Omega_{DT}(M) \propto M(M - M_P).$$

It follows that the modified entropy law is

$$S_{DT}(M) = 4\pi k_B \frac{G}{\hbar c} M(M - M_P), \qquad M \geq M_P,$$

and

$$S_{DT}(M) = 0, \qquad M < M_P.$$

This law has the correct macroscopic limit:

$$M(M - M_P) = M^2 - MM_P,$$

so

$$\frac{S_{DT}}{S_H} = 1 - \frac{M_P}{M},$$

and therefore

$$\lim_{M/M_P \to \infty} \frac{S_{DT}}{S_H} = 1.$$

Hence classical entropy is recovered asymptotically, while the admissible geometric state count collapses at the coherence boundary.

10 Modified Temperature and the Threshold Regime

The black-hole temperature is obtained from the thermodynamic relation

$$\frac{1}{T} = \frac{\partial S}{\partial E}, \qquad E = Mc^2.$$

Hence

$$\frac{1}{T} = \frac{1}{c^2}\frac{dS}{dM}.$$

Using

$$S_{DT}(M) = 4\pi k_B \frac{G}{\hbar c} M(M - M_P),$$

we obtain

$$\frac{dS_{DT}}{dM} = 4\pi k_B \frac{G}{\hbar c}(2M - M_P).$$

Therefore

$$\frac{1}{T_{DT}} = \frac{4\pi G k_B}{\hbar c^3}(2M - M_P),$$

and thus

$$T_{DT}(M) = \frac{\hbar c^3}{4\pi G k_B}\frac{1}{2M - M_P}, \qquad M \geq M_P.$$

This expression has several immediate consequences.

First, for $M \gg M_P$,

$$2M - M_P \sim 2M,$$

so

$$T_{DT}(M) \sim \frac{\hbar c^3}{8\pi G k_B M},$$

which is precisely the usual Hawking scaling in the large-mass limit.

Secondly, at the coherence boundary $M = M_P$, the temperature remains finite:

$$T_{DT}(M_P) = \frac{\hbar c^3}{4\pi G k_B M_P} = \frac{T_P}{4\pi},$$

where

$$T_P = \sqrt{\frac{\hbar c^5}{G k_B^2}}$$

is the Planck temperature.

Thus the threshold regime has a finite terminal temperature rather than a divergent one. The endpoint is not an ultraviolet blow-up. Instead, the geometric black-hole branch ends at a finite-temperature coherence boundary.

Remark 3. *The finiteness of $T_{DT}(M_P)$ does not imply continued radiative evolution below the boundary. Temperature remains an indicator of the boundary state's effective excitation scale, but the existence of a finite threshold temperature does not guarantee the existence of admissible lower-mass geometric microstates into which the object can radiate.*

This is the key distinction between the orthodox and Darke-theoretic endpoint pictures. In the orthodox picture, the nonzero temperature at low mass tends to drive continued and increasingly violent evaporation. In the present framework, that inference fails because admissible geometric phase space closes at the threshold.

11 Heat Capacity and Threshold Stability

The modified temperature law also changes the thermal response of the black-hole branch. Differentiating

$$T_{DT}(M) = \frac{\alpha}{2M - M_P}, \qquad \alpha := \frac{\hbar c^3}{4\pi G k_B},$$

gives

$$\frac{dT_{DT}}{dM} = -\frac{2\alpha}{(2M - M_P)^2}.$$

Hence the heat capacity

$$C_{DT} := \frac{dE}{dT} = c^2 \frac{dM}{dT}$$

is

$$C_{DT}(M) = -\frac{c^2}{2\alpha}(2M - M_P)^2 = -\frac{2\pi G k_B}{\hbar c}\,(2M - M_P)^2.$$

Several consequences follow immediately.

First, $C_{DT}(M) < 0$ for all $M \geq M_P$, so the geometric black-hole branch retains the familiar negative heat capacity of the Hawking regime.

Secondly, at the coherence boundary,

$$C_{DT}(M_P) = -\frac{2\pi G k_B}{\hbar c}\, M_P^2 = -2\pi k_B,$$

which is finite.

Thus the branch remains thermodynamically negative-capacity all the way to the threshold, but the pathology is softened: neither temperature nor heat capacity diverges. The branch reaches a finite terminal thermal state and then terminates because its admissible geometric continuation closes.

Remark 4. *The role of the heat-capacity calculation is diagnostic rather than foundational. It shows that the modified entropy law yields a threshold regime that remains thermodynamically well-defined up to the boundary, without the orthodox divergence.*

With the explicit definitions above, the physical content of the formation sector is now fixed. Density, curvature burden, and compressive convergence together define the local coherence burden,

$$b(x,t) = \alpha\,\hat{\rho}(x,t) + \beta\,\hat{K}(x,t) + \gamma\,\hat{\Theta}(x,t),$$

while the remaining local support for recoverable exterior geometry is

$$s(x,t) = \frac{1}{1 + b(x,t)}.$$

Accordingly, gravitational collapse drives the confinement functional upward both by increasing burden and by reducing support. Horizon formation then occurs when this coupled burden-support ratio reaches its critical value.

12 Evaporation Dynamics and the Coherence Boundary

We now derive the modified mass-loss equation. The standard power law for Hawking-like emission scales as

$$P \propto AT^4,$$

where A is horizon area and T is black-hole temperature. Using the classical area scaling

$$A(M) = \frac{16\pi G^2 M^2}{c^4},$$

together with the modified temperature

$$T_{DT}(M) = \frac{\hbar c^3}{4\pi G k_B} \frac{1}{2M - M_P},$$

one obtains

$$P_{DT}(M) \propto M^2 (2M - M_P)^{-4}.$$

Since

$$\frac{dM}{dt} = -\frac{P}{c^2},$$

the evaporation law takes the form

$$\frac{dM}{dt} = -\Gamma \frac{M^2}{(2M - M_P)^4}, \qquad M \geq M_P,$$

where $\Gamma > 0$ is an effective radiation constant that absorbs the usual numerical and species-counting factors.

This equation again has the correct large-mass behaviour. For $M \gg M_P$,

$$(2M - M_P)^4 \sim 16M^4,$$

so

$$\frac{dM}{dt} \sim -\frac{\Gamma}{16M^2},$$

which reproduces the usual Hawking scaling up to constant normalisation.

The threshold behaviour is more interesting. At $M = M_P$,

$$\left.\frac{dM}{dt}\right|_{M=M_P} = -\frac{\Gamma}{M_P^2},$$

which is finite. The rate does not diverge.

To understand the endpoint more clearly, set

$$M = M_P + \epsilon, \qquad \epsilon > 0.$$

Then

$$2M - M_P = M_P + 2\epsilon,$$

and

$$M^2 = (M_P + \epsilon)^2.$$

Hence near threshold,

$$\frac{d\epsilon}{dt} = -\Gamma \frac{(M_P + \epsilon)^2}{(M_P + 2\epsilon)^4}.$$

As $\epsilon \to 0^+$,

$$\frac{d\epsilon}{dt} \to -\frac{\Gamma}{M_P^2}.$$

Therefore the near-threshold behaviour is approximately linear:

$$\epsilon(t) \approx \epsilon_0 - \frac{\Gamma}{M_P^2} t.$$

It follows that the object reaches $M = M_P$ in finite time.

Proposition 1 (Finite-time threshold arrival)**.** *Suppose the black-hole mass evolves according to*

$$\frac{dM}{dt} = -\Gamma \frac{M^2}{(2M - M_P)^4}, \qquad M(0) > M_P,$$

with $\Gamma > 0$*. Then the mass decreases monotonically and reaches the coherence boundary* $M = M_P$ *in finite time.*

Proof. The right-hand side is strictly negative for $M > M_P$, so $M(t)$ is monotonically decreasing. Near $M = M_P$, the mass-loss rate approaches the finite negative constant $-\Gamma/M_P^2$. Hence the remaining mass gap $\epsilon = M - M_P$ decreases linearly to zero over finite time. □

This finite-time arrival is not yet the full endpoint statement. The stronger claim is that once the boundary is reached, the geometric black-hole branch has no admissible lower-mass continuation.

13 Why Radiation Terminates at the Boundary

A nonzero threshold temperature might appear to invite continued emission below M_P. The present framework rejects that continuation on structural grounds.

The reason is that evaporation is not driven by temperature alone. Radiation requires admissible target states. A radiative transition can occur only if the system possesses accessible lower-energy microstates within the same physical regime. Here the relevant regime is the geometric black-hole branch.

But by construction, the accessible geometric phase space collapses at the threshold:

$$\Omega_{DT}(M_P) = 0$$

in the sense that no lower-mass geometric microstates remain admissible below the boundary. Equivalently, the geometric validity condition

$$\Phi(M) = \frac{M}{M_P} \geq 1$$

fails for $M < M_P$.

Hence the correct interpretation of the endpoint is not that the object continues to evaporate indefinitely with a modified rate. Rather, the branch itself closes. There is no admissible lower-mass geometric state into which the object can radiatively transition while remaining a black hole.

Principle 3 (Coherence-boundary termination)**.** *Within the present framework, evaporation terminates at* $M = M_P$ *not because the effective threshold temperature vanishes, but because the geometric black-hole regime ceases to possess admissible lower-mass microstates below the coherence boundary.*

This is one of the most important conceptual moves in the paper. Temperature is an indicator of excitation tendency within a regime. It is not by itself permission for indefinite continuation of the regime. Regime continuation requires admissible support and admissible target states. Once those close, the branch ends.

This is exactly where the burden principle becomes decisive. Below the threshold, the support required to maintain geometric encoding as a black-hole regime is no longer admissible. The endpoint is therefore not merely kinematic. It is a burden-limit termination of the geometric branch.

14 The Terminal Boundary State

We may now identify the endpoint precisely.

Definition 12 (Terminal boundary state)**.** *The terminal boundary state is the final admissible Planck-scale state reached when the geometric black-hole branch terminates at*

$$M = M_P.$$

It is not a lower-mass geometric black hole, but the boundary closure of the evaporating confinement regime.

Several comments are important here.

First, the terminal boundary state is not introduced arbitrarily to repair divergence. It is structurally forced by the joint action of the burden principle and the coherence-boundary state-count collapse.

Secondly, the state is radiatively closed with respect to further evaporation of the geometric black-hole branch. This is the proper sense in which evaporation terminates. The object need not be interpreted as dynamically featureless in every possible deeper theory, but it is no longer an evaporating black hole in the geometric sense developed here.

Thirdly, the terminal state is best understood as a boundary object between two descriptive layers. From above M_P, it is the endpoint of the geometric black-hole regime. From below, no continuation of that same regime is structurally admissible.

15 Thermodynamic Interpretation

The thermodynamic meaning of the construction may now be stated more fully.

Entropy is not being treated here as vague disorder. It is the logarithm of accessible geometric phase space:

$$S = k_B \ln \Omega.$$

The key Darke-theoretic correction is that accessible phase space is not presumed to remain geometrically meaningful down to arbitrarily low mass. Instead, geometric accessibility is itself burden-limited and terminates at the coherence boundary.

This gives the horizon a dual interpretation. On the macroscopic side, it remains the external interface of the confinement regime. On the statistical side, it counts the accessible encoding layer of the black hole's geometric branch. Horizon entropy therefore measures not merely area in isolation, but the available geometric representability of the confined state.

The temperature law then reflects the sensitivity of this accessible geometric phase space to changes in mass. At large mass, the ordinary inverse-mass relation is recovered. Near threshold, the deformation in state count prevents divergence. Yet the fact that the temperature remains finite does not contradict termination, because phase-space closure removes admissible lower transitions.

Thus the endpoint becomes thermodynamically coherent:

1. entropy collapses to its terminal boundary value,
2. temperature remains finite,
3. heat capacity remains finite,
4. the geometric branch still terminates,
5. no divergent final explosion occurs.

This is a cleaner endpoint than the orthodox one because it distinguishes excitation scale from branch admissibility.

16 Relation Between Confinement and the Coherence Boundary

The present paper introduces two threshold quantities with distinct but structurally related roles:

$$\Lambda \quad \text{and} \quad \Phi.$$

It is useful to state their relation in the strongest form that the present framework justifies.

The confinement functional Λ is the regime-scale admissibility variable. It measures the burden-to-support ratio associated with maintaining recoverable exterior geometry during collapse:

$$\Lambda(r) = \frac{\mathcal{B}(B_r; \tau_r)}{\mathcal{S}(B_r; \tau_r)}.$$

When Λ reaches its critical value, the system can no longer sustain recoverable exterior geometry throughout the collapse region, and the structural horizon forms.

The coherence functional Φ is the terminal geometric admissibility variable:

$$\Phi(M) = \frac{M}{M_P}.$$

It measures whether the evaporating black-hole branch still possesses sufficient geometric coherence to remain an admissible black-hole regime at all. When Φ falls to unity, the geometric branch reaches its final admissible boundary.

These variables should not be read as expressing two disconnected principles. Rather, they represent two projections of the same underlying admissibility logic across different scales of the black-hole life cycle.

1. **Formation-scale admissibility.** At large and intermediate scales, the relevant question is whether confinement burden can be supported over the persistence horizon of collapse. This is measured by Λ.

2. **Termination-scale admissibility.** At the terminal evaporation scale, the relevant question is whether the geometric black-hole branch still possesses admissible state-space support at all. This is measured by Φ.

Thus Λ governs admissibility of *formation and persistence*, while Φ governs admissibility of *terminal geometric continuation*. In that sense, Φ may be understood as the minimal geometric-sector projection of the same burden-support principle encoded more generally by Λ.

The full life cycle of the black hole is therefore governed by one underlying structural logic expressed in two threshold forms:

1. **Formation threshold:**

$$\Lambda(r_h) = 1$$

with

$$\Lambda(r) > 1 \quad \text{for } r < r_h,$$

which initiates the confinement regime and horizon formation.

2. **Termination threshold:**

$$\Phi(M) = 1,$$

equivalently

$$M = M_P,$$

which closes the admissible geometric branch and halts evaporation.

The conceptual gain is important. The black hole is not born under one principle and terminated under another. It is born and terminated under one admissibility structure viewed at different levels: first as a burden-support threshold of confinement, and later as a coherence threshold of the residual geometric branch.

17 Observational and Conceptual Consequences

Although the present treatment is primarily conceptual and semi-formal, it still points toward a more definite regime-level test program. The framework does not predict large deviations in ordinary macroscopic black-hole phenomenology at high mass. Its clearest distinctions arise near structural thresholds: at formation in the interpretation of horizon onset, and at termination in the late-stage thermodynamic sector.

Late-stage thermodynamic behaviour

The sharpest empirical distinction of the present model lies in the terminal evaporation regime. In the orthodox picture, the endpoint tends toward divergent temperature and increasingly pathological behaviour unless further physics intervenes. In the present framework, the modified entropy law yields

$$T_{DT}(M) = \frac{\hbar c^3}{4\pi G k_B} \frac{1}{2M - M_P},$$

so the terminal temperature remains finite at $M = M_P$, and the corresponding heat capacity also remains finite. The model therefore predicts that any physically meaningful late-stage black-hole completion should avoid the orthodox divergent endpoint and instead approach a finite thermodynamic boundary state.

Planck-scale boundary remnants

Because the geometric branch terminates at the coherence boundary rather than evaporating through it, the framework predicts a terminal Planck-scale boundary state. This is not a conventional low-mass continuation of the same black-hole branch, but the final admissible state of that branch. If such objects have cosmological or relic significance, they would most naturally appear in scenarios involving primordial black holes or other early-universe compact populations.

Formation-side interpretation

At the formation side, the model recasts horizon onset as the point at which the burden of recoverable exterior geometry exceeds admissible support:

$$\Lambda(r_h) = 1.$$

This does not yet yield a directly measurable deviation from standard strong-gravity phenomenology, but it does impose a different structural interpretation of collapse. The horizon is no longer merely a causal surface. It is the boundary at which the exterior geometric branch ceases to remain supportable throughout the collapse region.

Information accessibility

The framework also distinguishes between collapse of accessible geometric microstates and total ontological erasure. Because the entropy sector counts only admissible geometric microstates, the closure of Ω_{DT} at the coherence boundary does not imply absolute destruction of closure-bound structural content. This suggests that any viable information theory of black holes should distinguish geometric accessibility from deeper structural retention.

Near-horizon and ringdown interpretation

The present paper does not derive explicit corrections to shadow structure, photon rings, or ringdown spectra. However, the reinterpretation of the horizon as a structural admissibility boundary rather than merely a causal surface suggests that future extensions of the framework

should look for threshold-sensitive signatures in near-horizon response and post-merger relaxation. Any such effects would likely be subtle and secondary to the large-mass external regime.

Regime priority

The most important practical point is therefore one of regime priority. The present model should not be judged primarily by whether it reproduces ordinary astrophysical black-hole behaviour, which it is designed largely to preserve. It should be judged by whether it provides a more coherent account of threshold behaviour: formation as admissibility crossing, interior singularity as regime breakdown, and evaporation endpoint as finite coherence-boundary termination.

18 Falsifiability and Regime Tests

A framework paper should still say what could, in principle, count against it. The present model therefore makes several regime-level commitments.

First, any successful completion of black-hole evaporation that requires a genuinely divergent terminal temperature would stand against the coherence-boundary construction developed here.

Secondly, any exact continuation of the geometric black-hole branch to arbitrarily small $M \ll M_P$ without phase-space closure would contradict the present $\Phi \geq 1$ admissibility condition.

Thirdly, if a deeper canonical derivation of the burden-support fields $b(x,t)$ and $s(x,t)$ were to show that no threshold of the form

$$\Lambda(r_h) = 1$$

can be consistently associated with horizon formation, then the formation-side functional proposed here would require revision.

Fourthly, if a viable information theory of black holes were to prove that collapse of accessible geometric microstates necessarily implies total ontological erasure, then the accessibility split introduced in the present paper would fail.

These are not ad hoc vulnerabilities. They are the natural points at which the framework binds itself.

19 Comparison with the Orthodox Picture

The differences from the standard picture may be summarised succinctly.

Aspect	Orthodox picture	Darke-theoretic picture
Black-hole identity	Strong-gravity object with horizon and singular interior	Terminal confinement regime governed by admissibility and closure
Horizon	Causal escape boundary	Boundary where coherence burden of exterior recoverability exceeds admissible support
Interior	Singularity or unresolved UV completion	Maximal closure-dominant confinement phase
Information	Paradox between loss and recovery	Closure-bound; accessibility basis-dependent
Entropy	Area law continued formally downward	Accessible geometric phase space with threshold collapse
Temperature	Diverges as $M \to 0$	Remains finite at the coherence boundary
Heat capacity	Divergent endpoint pathology	Finite negative value at the boundary
Endpoint	Divergent or UV-completed disappearance	Finite-time arrival at M_P and closure of the geometric black-hole branch

The point is not to deny the empirical strength of the orthodox external regime. It is to replace its most pathological conceptual continuations with a more structurally admissible account.

20 Discussion

The present framework is conservative in one sense and revisionary in another. It is conservative because it preserves the large-mass external phenomenology of black holes: horizon structure, thermodynamic behaviour, and Hawking-like scaling all emerge in the appropriate regime. It is revisionary because it changes the ontology underneath those behaviours.

The deepest shift is from unrestricted extrapolation to admissibility-governed persistence. This shift has several consequences.

First, singularities no longer enjoy default ontological status. Once one recognises that persistence requires support and that support is finite, a divergence in the old variables is no longer a reason to infer physical infinity. It is a reason to identify regime breakdown.

Secondly, the endpoint of evaporation ceases to be a numerical nuisance requiring arbitrary repair. The endpoint becomes a structurally intelligible branch termination. The black hole does not evaporate through the coherence boundary because the geometric branch itself has no admissible continuation there.

Thirdly, the burden principle gives black-hole persistence a more realistic structural footing. Black holes are not merely allowed because equations can be extended. They persist because the relevant regime remains supportable. They terminate when it no longer is.

Several deeper developments remain open. A more fundamental derivation of the burden and support laws from the broader Darke core would further strengthen the formation sector,

particularly by justifying the specific choice

$$b(x,t) = \alpha\hat{\rho}(x,t) + \beta\hat{K}(x,t) + \gamma\hat{\Theta}(x,t), \qquad s(x,t) = \frac{1}{1+b(x,t)},$$

as the canonical effective realisation of the burden-support structure. A more refined state-count model could be developed for the threshold regime, potentially including corrections beyond the minimal polynomial deformation used here. A more explicit bridge to horizon imaging, ringdown phenomenology, or remnant cosmology would also be valuable.

None of these open developments weaken the main result. Even in its present form, the framework shows that Darke Theory can host a black-hole picture in which formation, interior interpretation, and endpoint thermodynamics are all made structurally coherent.

21 Conclusion

We have constructed a standalone Darke-theoretic whitepaper for black holes by unifying confinement, structural admissibility, and the coherence boundary into a single framework.

The first major claim is ontological. A black hole is not treated primarily as a singularity object, but as a terminal confinement regime. Collapse increases coherence burden until the system can no longer support ordinary exterior geometric recoverability. The horizon therefore marks the structural boundary at which the burden of exterior recoverability exceeds admissible support.

The second major claim is interpretive. The interior is not a literal physical infinity. It is a maximal closure-dominant confinement phase in which the classical descriptive basis no longer remains structurally complete. Information is therefore not forced into absolute destruction, but may persist as closure-bound structure.

The third major claim is thermodynamic. By imposing a strict geometric coherence condition

$$\Phi(M) = \frac{M}{M_P} \geq 1,$$

and interpreting entropy as the logarithm of accessible geometric phase space, one obtains a modified entropy law, a finite threshold temperature, a finite boundary heat capacity, and a non-divergent evaporation equation. The mass reaches $M = M_P$ in finite time, and evaporation terminates there because no admissible lower-mass geometric microstates exist below the coherence boundary.

The overall result is a black-hole picture with a clear structural life cycle:

1. formation through burden-threshold crossing,
2. persistence as a closure-bound horizon regime,
3. termination at the Planck coherence boundary.

This replaces the orthodox singular and divergent endpoint picture with a burden-limited, structurally admissible confinement-boundary termination. In Darke Theory, the black hole is thus the terminal admissible form of extreme confinement: a closure-bound regime whose horizon marks the failure of exterior recoverability and whose evaporation ends when geometric coherence itself can no longer be borne.